

Hospital Management System Using Python

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Introduction :

The Hospital Management System is a web application developed to automate and simplify hospital record management. It is built using Python, Flask, SQLite, HTML, and CSS to maintain patient information in a structured manner.

The system allows users to manage patient records by adding, viewing, updating, and deleting data easily. It helps reduce manual effort, improves record accuracy, and provides an efficient digital approach for hospital operations.



Abstract:

The Hospital Management System is a web-based application developed using Python, Flask, SQLite, HTML, and CSS. It is designed to simplify hospital operations by maintaining patient records digitally. The system allows users to add, view, update, and delete patient information efficiently. It minimizes paperwork, improves data accuracy, and provides a simple and user-friendly interface. This project demonstrates the practical implementation of Python Full Stack Development concepts learned during the 45-day training program.



Objectives :

- ✓ Digitize Patient Records
- ✓ Reduce Paperwork
- ✓ Improve Efficiency
- ✓ Perform CRUD Operations
- ✓ Secure Data Storage

! Problem Statement :

Traditional hospital record management relies on paper-based documentation, which often leads to data loss, duplication, and time-consuming manual work. Managing patient information manually makes searching and updating records difficult. This project provides a digital solution that stores patient information efficiently using a database.



Technologies Used



Python



Flask



SQLite



HTML5



CSS3



JavaScript



Git & GitHub



VS Code



Django



Python Concepts Used :

Variables and Data Types

Operators

Conditional Statements

Loops

Functions

Exception Handling

File Handling

Object-Oriented Programming



Database :

Patients Table

- Patient ID
- Patient Name
- Age
- Gender
- Disease
- Phone Number
- Address

Features :

 Add Patient

 View Patient

 Update Patient

 Delete Patient

 Database Storage

 Search Records



Modules



Patient Management



Database Management



Web Interface



System Architecture :



Hospital Management System

 User



Web Browser (HTML/CSS)



HTTP Request/Response





Software & Hardware Requirements :

Software Requirements:

- Operating System: Windows 10/11
- Programming Language: Python
- Framework: Flask
- Database: SQLite
- Frontend: HTML, CSS
- IDE: Visual Studio Code

Hardware Requirements:

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 500 GB Hard Disk / SSD
- System: Laptop or Desktop Computer
- Internet Connection (for development and updates)

Advantages

-  Easy to Use
-  Fast Performance
-  Less Paperwork
-  Secure Data
-  Time Saving



Future Enhancements :



Doctor Management



Appointment Booking



Billing System



Login Authentication



Prescription Module



Email Notifications



Learning Outcomes :



Python Programming



Full Stack Development



Database Management



CRUD Operations



Git & GitHub



Conclusion :

The **Hospital Management System** was successfully developed using **Python, Flask, SQLite, HTML, and CSS**. The application provides an efficient way to manage patient records by performing CRUD (Create, Read, Update, Update, Delete) operations through a simple and user-friendly interface. It reduces manual paperwork, improves data accuracy, and makes patient information easy to manage. This project helped in understanding Python programming, web development, database management, and full-stack application development, making it a valuable academic project and a strong foundation for future real-world applications.



ID	Name	Age	Gender	Disease	Contact
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Thank You

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